

*The Petree College of Arts and Sciences provides a supportive, student-centered learning environment. Through an interdisciplinary curriculum that promotes active learning and individualized instruction, students are encouraged to think broadly, to find their passion, to collaborate with others, and to connect with the broader community.*

**Oklahoma City University**  
**Database Design and Management (CSCI 3613)**  
**Fall, 2025**

**Lecture: T, R 16:00 pm – 17:15 pm (Sarkeys Science & Math Center 104)**

**Instructor:** Dr. Chenguang Xu (Shine)

**Email:** shine.xu@okcu.edu

**Office hours:**

Tuesday: 9:00 am – 11:00 am

Thursday: 9:00 am – 11:00 am and 13:00 pm – 14:00 pm

Or by appointment

**Office Room Number:** Sarkeys Science & Math Center 209B

**Online Meeting:** Please reach out directly to me, I will send you a meeting invite for a call or video conference meeting.

**Course Description:**

In this course we will study the core fundamentals of database design and management, including database history, database architecture, relational models, SQL, normalization and the use of databases in application development.

**Course Goals and Objectives:**

By the end of this course students will have an understanding of database management systems, be able to write queries against relational databases, design new databases, and troubleshoot potential performance issues.

**Required Textbook:**

Database Systems: A Practical Approach to Design, Implementation, and Management(6e), Thomas Connolly and Carolyn Begg. ISBN-13: 978-0132943260

**Required Materials:**

A laptop is required for this class. You must be able to install the software that is required for the class. Required and recommended software will be cross-platform and usable on Windows, MacOS, or Linux.

**Minimum Required:**

Operating System: (PC Windows 8 or 8.1; MAC 10.10.5 or higher)

Core i5 Processor, 8GB RAM, 250 GB Hard Drive

Graphics Card: 512 MB or higher Video Memory

**Recommended:**

Operating System: (PC Windows 10 Home or Pro; MAC 10.10.5 or higher)

Core i7 Processor, 8GB RAM, 500 GB or higher Hard Drive

Graphics Card: 512 MB or higher Video Memory

**Learning Management System:**

Students must access and use OKCU's online learning platform, "D2L" (Desire to Learn; <https://ocuonline.okcu.edu>) for accessing and turning in course materials as instructed.

Announcement, assignment/lecture note, exam schedule will be posted on the site. And, also, completed assignment(s) and program(s) should be submitted through Desire2Learn.

**Grading:**

|            | Percentage |
|------------|------------|
| Attendance | 5%         |
| Homework   | 20%        |
| Exam 1     | 15%        |
| Exam 2     | 15%        |
| Project    | 20%        |
| Final      | 25%        |
| Total      | 100%       |

|            |      |
|------------|------|
| 92% - 100% | = A  |
| 90% - 91%  | = A- |
| 87% - 89%  | = B+ |
| 82% - 86%  | = B  |
| 80% - 81%  | = B- |
| 77% - 79%  | = C+ |
| 72% - 76%  | = C  |
| 70% - 71%  | = C- |
| 67% - 69%  | = D+ |
| 62% - 66%  | = D  |
| 60% - 61%  | = D- |
| 0% - 59%   | = F  |

**Tentative Lecture Schedule:**

| Week | Date  | Topic                                          | Homework   | Project                       |
|------|-------|------------------------------------------------|------------|-------------------------------|
| 1    | 08/26 | Syllabus                                       |            |                               |
|      | 08/28 | Introduction                                   |            |                               |
| 2    | 09/02 | Database Architecture                          |            |                               |
|      | 09/04 | Relational Model                               | Homework 1 |                               |
| 3    | 09/09 | <i>Project Plan</i>                            |            | <i>Project Plan Due</i>       |
|      | 09/11 | Relational Algebra                             |            |                               |
| 4    | 09/16 | Relational Algebra                             |            |                               |
|      | 09/18 | Relational Algebra<br><i>Project Breakdown</i> | Homework 2 | <i>Project Breakdown Due</i>  |
| 5    | 09/23 | <i>Project Breakdown</i>                       |            |                               |
|      | 09/25 | <i>Project Breakdown</i><br>Review for Exam 1  |            |                               |
| 6    | 09/30 | <b>Exam 1</b>                                  |            |                               |
|      | 10/02 | Exam review, Relational Calculus               | Homework 3 |                               |
| 7    | 10/07 | SQL Part 1                                     |            |                               |
|      | 10/09 | SQL Part 1, SQL Part 2                         | Homework 4 |                               |
| 8    | 10/14 | SQL Part 2                                     |            |                               |
|      | 10/16 | <i>Project Progress 1</i>                      |            | <i>Project Progress 1 Due</i> |
| 9    | 10/21 | <b>Fall Break</b>                              |            |                               |
|      | 10/23 | SQL Part 3                                     | Homework 5 |                               |
| 10   | 10/28 | Data Definition                                |            |                               |
|      | 10/30 | Views                                          | Homework 6 |                               |
| 11   | 11/04 | Transactions, Privilege                        |            |                               |
|      | 11/06 | Advanced SQL                                   | Homework 7 |                               |
| 12   | 11/11 | <i>Project Progress 2</i>                      |            | <i>Project Progress 2 Due</i> |
|      | 11/13 | <i>Project Progress 2, Review for Exam 2</i>   |            |                               |
| 13   | 11/18 | <b>Exam 2</b>                                  |            |                               |
|      | 11/20 | Entity Relationship Modeling                   |            |                               |
| 14   | 11/25 | Normalization                                  |            |                               |
|      | 11/27 | <b>Thanksgiving</b>                            |            |                               |
| 15   | 12/02 | Normalization                                  | Homework 8 |                               |
|      | 12/04 | Exam review, <i>Project Demo</i>               |            | <i>Project Demo Due</i>       |
| 16   | 12/09 | <i>Project Demo</i>                            |            | <i>Project Demo Due</i>       |
|      | 12/11 | <i>Project Demo, Review for Final Exam</i>     |            | <i>Project Report Due</i>     |
| 17   | 12/16 | <b>FINAL EXAM (4:00pm – 6:00pm)</b>            |            |                               |

**Attendance Policy / Makeup Work:**

Attendance will be taken at every class period and will be reported in accordance with university policy. You are *urged* to attend every lecture and are responsible for any material presented in your absence.

Make-up exams will only be given **at the discretion of the instructor**, with documentation from the student regarding the reason for missing the exam.

**Contingency Plans in Case of Campus Closure:**

In the case of a school closing during your scheduled class time due to inclement weather or emergency, your instructor will email you and provide an announcement in D2L regarding how class will proceed. This could include moving to virtual delivery and/or providing course content in D2L for online delivery. Due to the short duration of the graduate courses in this program, forgoing learning one week is not an option. Learning objectives will not be omitted.

**Incomplete Grade Policy:**

A grade of Incomplete (“I”) will only be assigned if a single exam cannot be completed due to documented extenuating circumstances such as hospitalization or a death in the family. The “I” will be removed in accordance with university policy stated in the 2018-19 web-based undergraduate catalog.

**Exams:**

All exams will include simple objective questions, “multiple-multiple” choice questions, pseudo code writing and short answer questions.

Students are responsible for all material covered in class and all reading and problem assignments. The best guide for exam preparation will be detailed lecture notes.

The unit exams will cover only material presented since the previous unit exam; the comprehensive final will cover all units equally.

**Projects/Programming Assignments:**

The grading of the Programming Assignments is a combination of completeness (all specifications are covered), correctness of results, and style. Students need to work on Projects as individuals and should NOT share code or copy other students work. All programming assignments are due at the beginning of class on the due date. I do not allow late work to be submitted unless there is prior written approval by me based on special circumstances.

**Late Work Policy:**

All work is submitted on time. Late work, unless previously approved by the instructor and for truly extenuating circumstances, will not be accepted.

**Cheating Policy:**

Cheating will not be tolerated, and cheating will be handled according to the university's academic honesty policy, stated in the 2022-23 web-based graduate catalog. Students should avoid any activity that might give the appearance of cheating, and students found cheating will receive a zero point grade on all relevant exams or quizzes.

**Electronics:**

Out of respect for the professor, fellow students, and class material, students should refrain from the use of electronic devices (for purposes unrelated to class) while in class.

**Syllabus Policy:**

The student is responsible for reading and understanding the policies outlined in the syllabus. The lecture schedule for the course is referred to as "tentative", yet every effort will be made to follow the schedule as closely as possible. The lecture topic will deviate from the schedule only when certain topics require more or less time than was scheduled. If a substantial change must be made to the course schedule (such as unavoidable cancellation of a class), students will be notified via university email and on D2L.

**Emergency:**

You will be notified through Blue Alert text, emails and web announcement by OKCU, and you can call 208-5911 from a campus phone or 911.

Sarkeys Science & Math building is for emergency shelter, and Activities Lawn is for exterior assembly location during this class meeting. For more shelter/assembly location on OKCU campus: <https://www.okcu.edu/main/emergency>.

**\* Tornado Shelter for Sarkey Building – any interior classroom on first floor;  
Evacuation Route for first floor of Sarkey Building – Northwest doors on first  
floor; assemble on activities lawn**

## **Academic Complaints:**

- Students are encouraged to utilize Oklahoma City University's internal complaint policies through the Office of the Provost/Vice President for Academic Affairs prior to filing a complaint with any external entity. Students may submit an official Academic Complaint at [Complaints - Oklahoma City University](#). Academic complaints submitted will be received by the Office of the Provost/Vice President for Academic Affairs and will be investigated by that office for action or further resolution.
  - Submitting an academic complaint via this form does not initiate a formal appeal process under the student handbook or the current catalogs found under quick links, <https://www.okcu.edu/academics/home>.
  - For information on submitting a complaint related to the accreditor of your program, please visit [Accreditation - Oklahoma City University](#).
  - For information on submitting a complaint about distance education courses, please visit [Distance Learning Complaints - Oklahoma City University](#).

## **Civil Rights, Equity, and Access:**

### **A. Nondiscrimination Policy**

- The Nondiscrimination Policy prohibits discrimination in the University community, including students, faculty, staff, guests, and visitors. Acts of discrimination, harassment based on a protected category, may limit, deprive, or deny educational/employment access, benefits, and opportunities. This policy prohibits discrimination based on race, ethnicity, national origin (including ancestry), religion, sex, gender identity, gender expression, sexual orientation, pregnancy, age, disability, domestic violence victim status, veteran, or military status, and dother protected categories under federal law. For a complete list of prohibited forms of discrimination, definitions, and examples, please visit University Policies Oklahoma City University to access the Nondiscrimination Policy.
- The University's Civil Rights Investigator, Dr. Kristi Pendleton, oversees prevention and response programs addressing discrimination in our community. To make a report, ask questions, or learn more, please contact Dr. Kristi Pendleton at (405) 208-6301 or at [kapendleton@okcu.edu](mailto:kapendleton@okcu.edu).

### **B. Sexual Harassment and Misconduct (Title IX)**

The Nondiscrimination Policy also prohibits all forms of sex discrimination, including sexual harassment, and stalking. That policy includes definitions and examples of these terms as well as the procedures for the investigation and resolution of Complaints. Many of these issues are also prohibited by Oklahoma criminal law and may be reported to campus or local police in addition to the University. When the University receives a report, the reporting party will be contacted by a Title IX administrator to discuss options for confidential support, support measures to address safety/access concerns, and formal and informal procedures for resolving Complaints as appropriate.

To make informed choices, it is important to be aware of mandatory reporting requirements for campus resources. Most University employees, including faculty, are required to report known/suspected

discrimination to the Compliance Coordinator within 24 hours. Confidential options including University Counseling, pastoral counseling, the health clinic, and advocacy services can maintain privacy. They do not disclose information about reports to university officials without explicit permission. Please visit [Title IX Information - Oklahoma City University](#) for more information about confidential resources, reporting options, policies and procedures, and contact information for University Title IX administrators. For emergencies, please call campus police at (405) 208-5911 or local police at 911.

### **C. Disability Accommodations (ADA / Section 504)**

If you believe that you are a student with a disability and need reasonable accommodations for a documented physical, mental, learning disability, attention disorder or another disability, please register with Disability and Access Services (DAS) by filling out the Student Application at the following link: [Disability & Access Services - Oklahoma City University \(okcu.edu\)](#). The DAS Staff is responsible for assessing student disability related accommodation requests. Reasonable accommodation may require early planning and is not provided retroactively, so please start this process as soon as possible. If you have general questions about reasonable accommodation, you may contact the DAS staff at (405) 208-5895 or by email at [das@okcu.edu](mailto:das@okcu.edu).

### **D. Religious Accommodation**

Oklahoma City University seeks to be supportive of religious observance among the members of our diverse campus community and to be as accommodating as possible. Students should discuss with their instructor at the beginning of the semester forms of religious observance (dress, fasting, specific prayer times) that may affect their full participation in the course. Students should also compare the class schedule to their own religious calendar to determine if there will be any class days in which the student expects to be absent due to the observance of a religious holiday. Students must notify the instructor, in writing, of the expected absence within the first two weeks of the semester. The instructor will then work with the student to develop a plan to reschedule any exams, assignments, or course activities for that day. The instructor, at his/her own discretion, will provide reasonable accommodation wherever possible. Students should recognize, however, that there may be some course aspects that cannot be rescheduled or accommodated, and it will therefore rest upon the student to determine whether they wish to remain enrolled in the course or have their grade potentially affected.